Towards Realistic UAV Vision-Language Navigation

Platform, Benchmark, and Methodology

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Project page

prince687028.github.io/OpenUAV

01 MOTIVATION

UAV object search is still flown as waypoints. A human translates “what to find” into “how to fly” before every mission.

Ground VLN assumes discrete moves on a graph. A drone moves continuously in 6 DoF, with inertia, attitude and obstacle constraints.

Simplified simulators hide the gap. Policies that pass

> **We need a platform, a benchmark, and a method built for continuous flight.**

02 WHY BENCHMARKS FALL SHORT

Action space. 2–4 DoF discrete actions cannot express continuous flight.

Dynamics. Simplified kinematics hide sim-to-real drift.

Scale. Few datasets cover many objects, scenes and instruction styles.

Dataset	DoF	Kin.	#Traj	Vocab	Actions	Environment
R2R	–	✗	7,189	3.1K	5	Matterport3D
TouchDown	2	✗	9,326	5K	35	Street View
AerialVLN	4	✗	8,446	4.5K	204	AirSim
CityNav	4	✗	32,637	6.6K	–	SensatUrban
Ours	6	✓	12,149	10.8K	264	OpenUAV

Table 1. Existing VLN datasets vs. ours (6 DoF, real kinematics).

03 OPENUAV PLATFORM

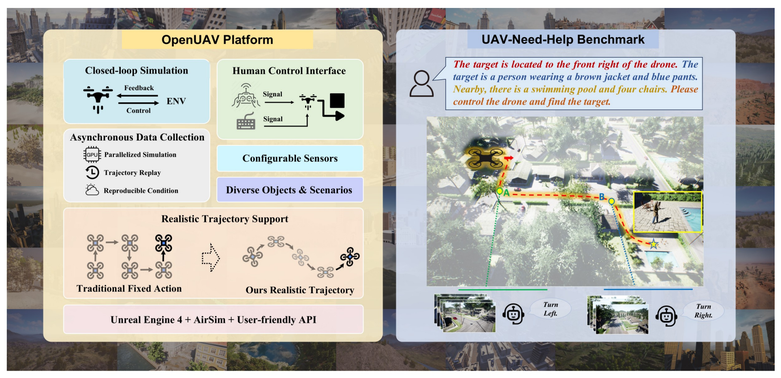


Figure 1. Closed-loop simulation, human control interface, configurable sensors and trajectory replay.

Unreal Engine 4 + AirSim, 22 high-fidelity scenes with diverse objects and lighting.

Closed-loop, parallelized simulation with reproducible conditions.

Human control interface records continuous flights; sensors are collected asynchronously.

> **One platform, one benchmark, one policy — all built for continuous 6-DoF flight.**

04 CONTRIBUTIONS

1 OpenUAV. A realistic, closed-loop UAV simulation platform with a human control interface.

2 UAV-Need-Help. A 12k-trajectory, 6-DoF benchmark with three levels of assistant guidance.

3 Hierarchical navigation LLM. Far target pose, then fine trajectory, trained with backtracking data aggregation.

05 DATA COLLECTION

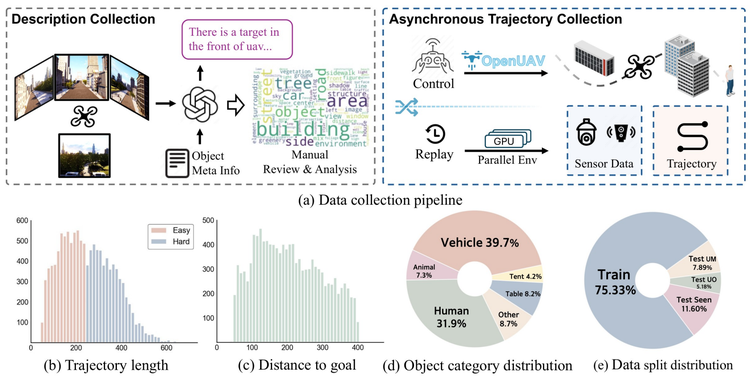


Figure 2. (a) Instruction vocabulary, (b–d) trajectory and target statistics, (e) split design.

Human pilots fly continuous trajectories; UAV states are recorded and sensor data is collected asynchronously.

GPT-4 descriptions are generated for each target and manually checked.

Each trajectory ends within 5 m of the target; collisions are discarded.

06 DATASET AT A GLANCE

12,149 trajectories

6 DoF continuous flight

89 object categories

22 scenes

Split	Traj.	Objects	Scenes
Train	9,152	76	20
Test Seen	1,410	seen	seen
Test Unseen Map	958	seen	2 unseen
Test Unseen Object	629	13 unseen	seen

Easy < 250 m · Hard > 250 m · target distance 50–400 m

07 UAV-NEED-HELP BENCHMARK

The UAV searches from the target description, the scene, and guidance from an assistant.

L1 Full guidance
Continuous guidance to the target throughout the episode.

L2 Coarse guidance
Guidance only at critical states, when the policy is uncertain.

L3 Avoidance
Warns about collisions and hazards only.

08 WHAT THE INSTRUCTIONS LOOK

> *“The target is located to the front right of the drone. The target is a person wearing a brown jacket and blue pants. Nearby, there is a swimming pool and four chairs. Please control the drone and find the target.”*

Target descriptions combine direction, appearance, and surrounding context.

09 METHOD: HIERARCHICAL

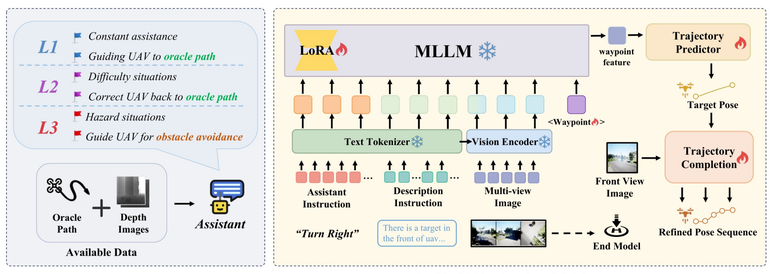


Figure 3. Instruction and five views are encoded by an MLLM; a far target pose is decoded first, then a fine 6-DoF trajectory.

Understand. An MLLM fuses the instruction with five RGB views and produces a task representation.

Decide far, then near. The model predicts a far target pose, then a fine trajectory in continuous action space.

Learn from backtracking. Data aggregation samples recovery trajectories, improving obstacle avoidance and robustness.

> **Why far-to-near?** Direct regression must cover a huge continuous search space. A far target pose narrows down where to go; the fine trajectory covers a short horizon, and the policy re-plans from new views.

10 RESULTS: TEST SEEN (L1)

2.1×

success rate over the CMA baseline

17.45% vs 8.37% · same protocol

Method	NE ↓	SR ↑	OSR ↑	SPL ↑
CMA	135.73	8.37	18.72	7.90
Ours	106.28	16.10	44.26	14.30
Ours-DA	98.66	17.45	48.87	15.76
Human	14.15	94.51	94.51	77.84

Table 2. Test Seen, L1 assistant. Simulation results; real flights remain subject to airspace and site conditions.

11 GENERALIZATION

22.42%
SR on unseen objects
(CMA: 9.06%)

4.18%
SR on unseen maps
(CMA: 2.30%)

The same policy transfers to new maps and new object categories without retraining.

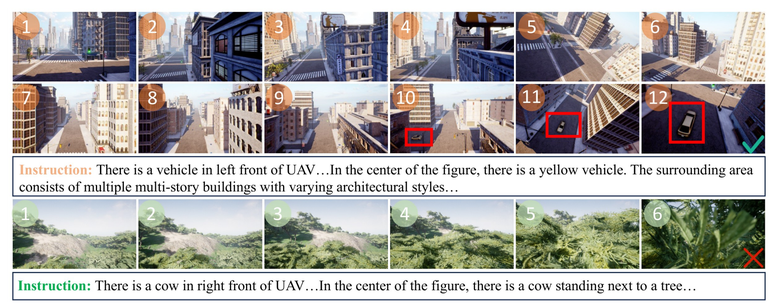


Figure 4. Closed-loop rollouts in held-out scenes; when the target is ambiguous, the policy re-observes.